Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/14/2013
PI Name:	Midura, Ronald J Ph.D.		
Project Title:	Hypogravity induces vascular atrophy leading to bone loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44195	**Congressional District:**	11
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	02/01/2010	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle		
Flight Assignment:	STS-135 NOTE: Extended to 1/31/2014 per NSSC/A. Chu-ARC (Ed., 2/21/2013)		
Key Personnel Changes/Previous PI:	Dr. N. Patrick McCabe, the senior research investigator on the project, left the research program in mid-2012 and is no longer working on the project.		
COI Name (Institution):	Parsons-Wingerter, Patricia (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AE40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Hypogravity leads to a loss of ground reaction forces transmitted through the lower appendicular and axial skeleton. Loss of weight bearing (WB) loads leads to a state of progressive osteopenia in these skeletal sites. Hypogravity mediated unloading reduces mechanical compression forces resulting in decreased local bone strains and reductions in hydrodynamic forces within bone tissue. In addition to this mechanism, we hypothesize hypogravity mediates reductions in the vascular supply serving these bone sites that contribute to hypogravity induced osteopenia and may help explain the prolonged time to recover bone mineral density (BMD) and bone strength upon return to Earth. Vasculature associated with bone tissue will be assessed using 3D micro-angiography imaging, which uses contrast agents to fill blood vessels including those in bone followed by micro-CT imaging and histomorphometry. Using micro-angiography, we have developed the tissue and image processing means to assess the bone-associated vascular volumes in the hind limbs of rats. Using the hind limb unloaded (HLU) analog, a time-dependent loss of vascular volume in the lower legs will be monitored and compared to the temporal pattern of progressive bone loss at these skeletal sites. Blood flow and vascular volumes will be assessed in these rat groups and a subsequent effect of these treatments on bone volume, BMD, and vascular morphology in the tibiae and fibulae will be determined. A cause & effect relationship between blood supply and osteopenia will need to be investigated in a future study by directly modulating blood supply to the hind limbs of rats. The blood supply to the hind limbs of WB rats could be reduced using unilateral femoral artery ligation and treatment with a VEGF inhibitor. The blood supply to the hind limbs of HLU rats could be enhanced using a hydrogel-based timed release of VEGF in proximity to the tibia and fibula. In addtion, a loss of bone-associated blood supply could hamper a proper angiogenic response thereby delaying fracture healing. In this case, after HLU exposure, fibulae would be fractured and the extent of repair would be monitored by micro-CT imaging, followed by vascular volume assessments at the fracture callus site using 3D micro-angiography. Vascular volume, bone volume, BMD and vascular morphology within the fracture callus would be compared between HLU and WB rats. This proposal frames a novel hypothesis that could contribute new knowledge impacting on both NASA's Risk of Accelerated Osteoporosis and its risk of Impaired Fracture Healing. If hypogravity unloading leads to an atrophy of bone-associated vasculature, then a negative impact could result contributing to the progressive osteopenia observed during space flight. A reduced bone vascular supply will hamper nutrient and waste product exchange rates in cortical bone thereby stressing osteocytes metabolically, and quite possibly leading to some osteocyte death and localized osteonecrosis. Given the slow remodeling rate of cortical bone, such a possibility could manifest in long term skeletal health complications. First, the presence of any localized osteonecrosis within cortical bone may elevate the risk of fracture upon return to Earth after a long space mission. Second, the remodeling & revascularization of cortical bone in skeletally mature adults takes a long time, and this slow process may help explain why the recovery rates for BMD and bone strength are slow in astronauts returning from long duration space missions. A slow recovery of bone-associated vascular supply may also compound into long term health effects to astronauts in their later years of life (e.g., peripheral artery disease).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our current knowledge base suggests that bone healing is impaired during and after an extended exposure to simulated weightlessness. This study frames a novel hypothesis that states that hypogravity mediated unloading leads to a progressive atrophy of the vasculature supplying lower leg bones. In turn, this regression in bone-associated blood supply and vascular bed volume will have a negative effect on bone tissues resulting in additional and progressive osteopenia beyond that attributed to reductions in local bone strain. Should this association between vascular regression and deficient bone healing hold true, then these studies would manifest relevance to NASA’s existing manned space missions. In addition, these studies may also exhibit importance regarding the underlying causes of impaired bone healing in patients experiencing paralysis, immobility or extended bed rest here on Earth. Finally, this work may also exhibit relevance to other pathologies resulting in the poor healing of recalcitrant bone fractures here on Earth.
	This progress report represents the final year of funding for the Midura lab study: “Hypogravity induces vascular atrophy leading to bone loss”. In the year since the last progress report much has been completed as the project approaches its final phase. This progress report will include an update on the mouse specimens flown aboard the STS-135 space shuttle mission (CBTM-03 payload), which was a supplement to NNX10AE40G. The relatedness of this spaceflight study compared with the ground based rat study should allow the two to be analyzed similarly and compared to one another. In section A the original grant’s ground based rat study will have its task progress reported. Section B will cover the progress with the STS-135 samples and section C will cover the tasks that will be completed in the final months of project funding under an approved no-cost extension (NCE). A. Ground based studies 1) All animal based work has been completed, including all vascular profusions and tissue retrieval from specimens. 2) The tissue sections that were previously being used for immunohistochemical analysis have been shown to be just as efficiently imaged using light microscopy. Barium-perfused vessels show enough contrast with the tissue to be easily identified and scored. As such, tissue sections of perfused animals were imaged using light microscopy and full tissue section images were compiled. These tissue sections are still imaged for alpha-Smooth Muscle Actin (aSMA) antibody detection under fluorescent microscopy. The aSMA staining acts to verify the presence of vessels ensuring a minimal number of false positive vessels. These tissue sections are awaiting analysis using ImagePro software. Analysis of 2D histomorphometry will allow for vessel metrics to be quantified, these include vessel size and vessel number. 3) The micro-CT imaging of perfused rat specimens was completed and the raw data is currently undergoing 3D morphometric analysis. These analyses are currently being validated for consistency and calibration in order to produce valid results that could be compared with existing vascular literature. 4) 3D volumes of perfused rats were aligned to each animal’s baseline scan due to each animal’s anatomically unique characteristics. 3D volumes of perfused hind limbs were cropped to cover a 5-mm vertical region of interest (ROI) within the middle of the lower hind limb. These crops were then divided into 3 additional regions of interest classified as anterior, central, and posterior based on the anatomical position in the hind limb. Once the vessels and bone had been aligned and cropped, the bones were masked out of the image this prevents the bone from interfering with the analysis of vasculature.

Task Progress:	B. Spaceflight Study 1) The genetic analysis of all STS-135 (CBTM-03) spaceflight and synchronous ground-based control samples was completed. These specimens represent a similar mid-calf ROI as the prior mentioned rat HLU study but only for the left hind limb of untreated control mice in this study. These quantitative gene expression analyses were accomplished using a Qiagen qRT-PCR arrays interrogating the activity of 88 genes related to endothelial cell function. The genetic data obtained from these arrays was analyzed using 7500 systems software and Qiagen supplied analysis software. 2) Additional specimens for an HLU ground control study were obtained from Dr. Bouxseim's group at Harvard University. These samples included both HLU and WB control C57Bl/6 female mice (identical to those flown on STS-135). Specimens in the HLU group were exposed to HLU for the same duration as the space flight duration of the STS-135 mission (12.5 days). The control mice in this study were the same age as the HLU experimental mice and all animals were treated similarly to those from the STS-135 mission. The mid-calf ROI of each animal's left hind limb was utilized for this control study and (given the small size of the C57Bl/6 female mice used) the bone and skeletal muscle were not separated. The proximal end was placed in RNAlater at 4° C to preserve RNA for genetic analysis of factors associated with endothelial cell alterations. The distal end was placed in 10% neutral-buffered formalin and stored at 4°C. The same analysis of gene expression was done using Qiagen gene arrays. This genetic data was also analyzed statistically using 7500 system software and Qiagen supplied analysis software. The distal ends were decalcified and embedded in tissue freezing medium and were cryo-sectioned for immunohistochemical analysis. 3) The genetic analysis of the spaceflight and Harvard studies was divided into 4 unique groups; spaceflight ground control, spaceflight, Harvard ground control, and Harvard HLU. These groups were designated to allow for comparison of control groups as well as comparison of the spaceflight animals with the HLU mice. These findings are of great importance as hind limb unloading is a NASA approved model for spaceflight studies. 4) It was determined that the von Willebrand factor (vWF) staining was too variable to be of use in the histomorphometric analysis of vasculature. As such it was found that isolectin B4 (IB4) gave much more consistent and useful staining, especially of microvasculature (vessel diameters less than 50 microns). 5) The immunohistochemical staining done on the tissue sections utilized fluorescently tagged primary antibodies alpha-Smooth Muscle Actin (aSMA) and Isolectin B4 (IB4). The tissue sections were stained along with DAPI which fluorescently labeled the nuclei. The tissue sections were then imaged on a fluorescent microscope and full-sized tissue section images were compiled. These immunohistochemical sections are undergoing visual validation and morphometric analysis using ImagePro software. Analysis of vasculature will quantify vessel size and vessel number in several regions of interest throughout the tissue. C. Task progress for the remaining time of funding Remaining slides from the 2D histomorphometry for both the ground based and space based studies will be scanned in the next 12 weeks using the newly identified procedures. The 2D analysis of these tissue sections will be completed in the following 12 weeks as the analysis software is completed by Image IQ. Thus, it will take my research team another 5-6 months to complete the entire work-plan for NNX10AE40G. The 3D morphometric analysis of vascular perfusions carried out in the ground based rat studies will be completed within this timeframe as the methodology of analysis becomes standardized and validated. The final report will be compiled including data from the (1) ground based rat study, (2) STS-135 spaceflight study, and (3) Harvard HLU mouse study to compare to the spaceflight study.
Bibliography Type:	Description: (Last Updated: 03/01/2017)
Abstracts for Journals and Proceedings	Androjna C, McCabe NP, Midura R. "Alterations in Factors Associated with Endothelial Cell Function in the Hind Limbs of Mice flown aboard STS-135." 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. , Jun-2012
Articles in Peer-reviewed Journals	Androjna C, McCabe NP, Cavanagh PR, Midura RJ. "Effects of spaceflight and skeletal unloading on bone fracture healing." Clinical Reviews in Bone and Mineral Metabolism. 2012 Jun;10(2):61-70. http://dx.doi.org/10.1007/s12018-011-9080-z , Jun-2012